McGill University
Department of Mathematics and Statistics
MATH 598 Topics in Probability and Statistics

Student Name: Wen, Zehai Student ID# 260956530

Student Email: zehai.wen@mail.mcgill.ca

Degree Program and Year: PhD in Statistics Term: Fall 202609 Winter

Topics to be covered: (Outline the Main reading material.)

Topic Title: Uncertainty Quantification for Machine Learning

This course provides a formal introduction to the conformal prediction framework, a model-agnostic uncertainty quantification method applicable to machine learning and artificial intelligence systems. The curriculum examines the mathematical mechanisms of distribution-free inference and their computational implementation. Students will derive finite-sample, model-agnostic coverage bounds utilizing exchangeability and permutation tests. The course connects theoretical foundations with applied methodology, analyzing the use of conformal inference in high-dimensional settings, trustworthy AI pipelines, computational biology, and drug discovery.

Method of Evaluation: Homework 30%, Participation 10%, Team project 40%, Presentation 20%

Student Signature and Date: July 8, 2026

Instructor Name: Archer Yang Signature and Date: June 1, 2026 Archer Yang

For Undergraduate Students - Chief Advisor Approval:

Professor Djivede Kelome Signature and Date: June 1, 2026

For Graduate Students – Graduate Program Director Approval

Professor Brent Pym Signature and Date: June 1, 2026

Please return form with all 3 signature via email to the Undergraduate Program Coordinator, Angela White at angela.white@mcgill.ca

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